Applied Clinical Informatics © 2022. The Author(s).



## Supplementary Appendix

**Appendix Table A** Description of recommended eHealth usability evaluation methods specifying the stage of software lifecycle (such as requirements engineering, design, and evaluation) and associated equivalent terms that can be used synonymously with each other

| Recommended eHealth usability evaluation method | Stage of software lifecycle | Description of recommended eHealth usability evaluation method | Equivalent terms | Quotes |
|---|---|---|---|---|
| Remote User Testing (combined with Think Aloud or Interview) | Evaluation[64] | Characterized by a decentralized test implementation.[21] Evaluator and test participants are located at different physical locations.[21] | Remote Usability Testing[65]; Remote Control Evaluation[64] | "Well, Remote User Testing is basically just a change of location." The simplified "… technical environment, working at distance …" are beneficial. "The effort for pre- and post-processing of Remote User Testing remains the same. However, Remote User Testing is beneficial for evaluating eHealth systems quickly due to its simplified technical framework, such as conducting the evaluation at distance." "Combining Remote User Testing with Think Aloud or Interview is recommendable, since insights into participants' thought processes are difficult to gain solely from observations when tasks are performed." |
| Expert Review (Walkthrough or Talkthrough) | Evaluation[35] | Experts mentally or verbally assess the usability[66] of an eHealth system. | Usability Expert Evaluations[35]; Evaluation by Experts[35] | "To keep the effort low, an Expert Review is always a quick choice to accomplish a usability evaluation in health care. If both usability and domain experts from the respective medical domain are available, it is advisable to carry out a Walkthrough or Talkthrough." |
| Rapid Iterative Test and Evaluation Method | Evaluation[35] | Data preparation and analysis is performed after each test participant has performed the evaluation or not later than an evaluation day.[67] | – | The Rapid Iterative Test and Evaluation Method lends itself to "… quickly involve future users, which is an important precondition to develop user friendly eHealth systems." |
| Heuristic Evaluation | Evaluation[35]; Early system design stage[24] | Experts use heuristics to assess the usability[49] of eHealth systems. | – | "If a usability evaluation should be carried out very quickly and if there are good relationships with usability professionals, I can imagine that Heuristic Evaluation will be implemented within one or two weeks, which is very beneficial for the development of eHealth systems." |
| Pluralistic Walkthrough | Design[49] | Usability experts, software developers, and test participants jointly evaluate the usability[49] of an eHealth system. A central aspect of Pluralistic Walkthrough is integrating test participants who are representative of prospective users.[68] | – | "Developers, usability engineers, and medical computer scientists, you can obviously use them to implement a Pluralistic Walkthrough in the context of health care."<br>"One opportunity to save effort is that developers, usability engineers, and medical computer scientists jointly prepare the analysis of the evaluation." |
| Agile User Test | Evaluation[66] | In the course of an Agile User Test, test participants who are representative of the target | – | An Agile User Test "… means having shorter tests performed more often; with fewer test participants and shorter tasks, or just one task, |

**Appendix Table A** (*Continued*)

| Recommended eHealth usability evaluation method | Stage of software lifecycle | Description of recommended eHealth usability evaluation method | Equivalent terms | Quotes |
|---|---|---|---|---|
| | | group perform tasks to evaluate an eHealth system.[66] | | even in combination with an Expert Review. Then it is possible to react quickly to usability findings that are important for eHealth systems that should be rapidly deployed." |
| Guerrilla Testing | Evaluation[49] | Usually conducted in a public place with randomly selected test participants.[69] | Discount Usability Engineering[49] | "It's possible with fitness apps compared to safety-critical eHealth systems." The realization of Guerrilla Testing is "… rather difficult with mHealth systems which are used for personal health care because you rarely find the right target group on site." |
| Feature Inspection | Design[49] | Functionalities of an eHealth system are evaluated with a feature checklist to examine typical action processes.[49] | – | "During Feature Inspection, I check whether the system meets the usability requirements. That is beneficial to avoid additional costs and to prevent delays concerning the development of eHealth systems." |
| A/B Testing | As soon as a prototype is available[70] | A/B Testing is preferably used to evaluate web applications to compare different versions of the user interface with different groups of test participants.[70] | Automated A/B Usability Testing and Improvement Method[70] | "Currently, live A/B Testing is used in the digital sector, which is cool because you can combine qualitative with quantitative data. This is a great opportunity to evaluate eHealth systems such as patient portals." |
| Crowd Testing | Evaluation[28] | Web-based applications are tested quickly, in real-time, and with real-time users.[71] | Crowdsourced Software Testing[72]; Crowdsourced Testing[28] | "Crowd Testing are surveys, so called quantitative methods. The evaluations are implemented automatically under real conditions, which is useful to evaluate eHealth systems aimed at patients. It's important to consider the representativeness of test participants. This is especially important if patients or chronically ill persons are the prospective users." |

Note: Recommended eHealth usability evaluation methods are ordered by the number of experts' choice.

Applied Clinical Informatics © 2022. The Author(s).

**Supplementary Appendix Table B** Description of potentially useful eHealth usability evaluation methods specifying the stage of software lifecycle (such as requirements engineering, design, and evaluation) and associated equivalent terms that can be used synonymously with each other

| Potentially useful eHealth usability evaluation method | Stage of software lifecycle | Description of potentially useful eHealth usability evaluation method | Equivalent terms |
|---|---|---|---|
| Assessing Cognitive Workload | Evaluation[62] | Quantification of participants' mental effort in the context of using a prototype or software by means of a Questionnaire.[62] | – |
| Cognitive Walkthrough | Requirements and Evaluation[35]; Early Design Stage[24] | By means of a Cognitive Walkthrough, an expert mentally records the interactions of prospective users step by step.[24] | Evaluation |
| Collaborative Walkthrough | Design[65] | Test participants are guided during the recorded evaluation by the evaluator if there are uncertainties.[65] | Remote Collaborative Walkthrough[65] |
| Consistency Inspection | Design or early in the software lifecycle[49] | A Consistency Inspection is comparable to a Pluralistic Walkthrough, a so called group inspection method,[49] but in the course of this several projects will be evaluated simultaneously to identify similarities.[49] | – |
| Cooperative Usability Testing | Evaluation[63] | Collaboration of users and usability experts in the course of a usability evaluation to accomplish a combination of domain expertise and human computer interaction expertise.[63] | – |
| Evaluation Workshop | Evaluation[62] | An Evaluation Workshop takes place between users and software developers, in the course of which test participants evaluate a system by means of the performance of usability assessment tasks.[62] | – |
| Formal Usability Inspection | Design or early in the software lifecycle[49] | A Formal Usability Inspection consists of a multi-step process that combines a Heuristic Evaluation with a simplified version of a Cognitive Walkthrough.[49] | – |
| Group Usability Testing | Late design stage; Evaluation[73] | During Group Usability Testing, a group of users simultaneously, but independently, perform usability assessment tasks[73] of an eHealth system under the supervision of evaluators. | – |
| Guideline Review | Evaluation[66] | A Guideline Review is the addition of a classical inspection to a Heuristic Evaluation, in the course of which the usability of a system is evaluated against a set of usability guidelines.[66] | – |
| Heuristic Walkthrough | Evaluation[74] | Experts apply both heuristics and "thought-focusing" questions to identify usability issues.[74] | – |
| Informal Cognitive Walkthrough | Evaluation[32] | Composed of several Simplified Streamlined Cognitive Walkthroughs and Simplified Pluralistic Walkthroughs.[32] The Simplified Streamlined Cognitive Walkthrough reduces the preparation time suitable for the application in agile software development.[32] The approach of a Simplified Pluralistic Walkthrough consists of two Walkthroughs to evaluate a system, deployable with four representatives of customers and simplified documentation.[32] | – |
| Laboratory Usability Testing | Evaluation[35] | Most often, a usability test is conducted in a usability laboratory,[21] which is equipped with extensive video and audio technology. | – |
| Micro Tests | Micro Tests can be applied in the course of agile software development[75] | Micro Tests are defined as a limited number of online, Unmoderated Usability Tests, usually three to five usability tests with one to two tasks feasible for each test.[75] | – |

Applied Clinical Informatics © 2022. The Author(s).

**Supplementary Appendix Table B** (*Continued*)

| Potentially useful eHealth usability evaluation method | Stage of software lifecycle | Description of potentially useful eHealth usability evaluation method | Equivalent terms |
|---|---|---|---|
| Peer Tutoring | Evaluation[76] | eHealth usability evaluation method that specifically targets children as prospective users. Peer Tutoring means that children teach other children how to use the software being evaluated.[76] | – |
| Persona-Based Rapid Usability Kick-Off | Early-stage product development[77] | Half-day workshop with participants of the project team, such as software developers and usability experts.[77] | – |
| Perspective-Based Inspection | Design or early in the software lifecycle[49] | Different perspectives reveal different usability issues of a system.[68] | – |
| Rapid Usability Testing | Development[16] | User interactions are recorded during Rapid Usability Testing.[16] | – |
| Simplified Think Aloud | Evaluation[49] | Interview technique[31] that differs from traditional Thinking Aloud in the reduced number of test participants and the simplified version of documentation and dispensed video recording.[49] | – |
| Standards Inspection | Design or early in the software lifecycle[49] | A Standards Inspection means that an expert reviews the usability requirements of a software against system and usability standards.[66] | – |
| Streamlined Cognitive Walkthrough | Evaluation[32] | The Streamlined Cognitive Walkthrough is an approach in the course of which the number of questions asked to test participants for step-by-step evaluation in the course of a mental think-through of the usability of a software and the targeted documentation of usability problems were reduced.[32] | Simplified Cognitive Walkthrough[32] |
| Synchronous Usability Testing | Evaluation | Moderated in real time by an evaluator and conducted with a test participant who is in a different physical location, i.e., remote.[21] | – |
| Think Aloud | Evaluation[35]; Design, Implementation[24] | Test participants who are representative of prospective users perform tasks by expressing their thoughts aloud.[24] | Think-Aloud Protocol,[53] Concurrent Think Aloud,[53] Think-Aloud Tests[63] |

Note: Potentially useful eHealth usability evaluation methods are ordered alphabetically.

Applied Clinical Informatics © 2022. The Author(s).

**Supplementary Appendix Table C** Description of not recommended eHealth usability evaluation methods specifying the stage of software lifecycle (such as requirements engineering, design, and evaluation) and associated equivalent terms that can be used synonymously with each other

| Not recommended eHealth usability evaluation method | Stage of software lifecycle | Description of not recommended eHealth usability evaluation method | Equivalent terms | Quotes |
|---|---|---|---|---|
| Retrospective Testing (combined with Think Aloud or Eye Tracking) | Late design stage; Evaluation[54] | Test participants view the tasks they performed via video recording after completing the evaluation and speak their thoughts out loud.[54] | Retrospective Think-Aloud Method,[54] Retrospective Think Aloud Usability Testing[53] | "These retrospective methods already have a benefit, but the benefit is bought at great costs. That's my overall assessment."<br>"During Retrospective Testing, the whole evaluation is done twice, which is obviously a high-quality level but with a greater effort in comparison to a traditional usability evaluation."<br>"With time and money constraints, I would not recommend Retrospective Testing to evaluate eHealth systems."<br>Furthermore, the deployment of Eye Tracking is not recommendable because "Eye Tracking is always time-consuming." |
| Focus Group | Requirements[78] | A group discussion is usually conducted at the beginning, followed by a joint evaluation of the usability of an eHealth system by small groups of two-three test participants and concluded by a discussion among all participants.[79] | – | "As I have to keep the level of effort low, I wouldn't go with Focus Groups with completely different persons either because the effort in implementation and preparation is quite high."<br>"I would never seriously evaluate anything with Focus Groups, to be honest. Focus Groups do not belong to evaluation methods either and are thus not recommendable to evaluate eHealth systems. Focus Groups belong to user research."<br>"I have to think about guiding questions for the Focus Group, about moderation concepts or questions to be answered, otherwise it wouldn't be a Focus Group. This results in a high effort. Otherwise, it would simply become an open discussion." |
| Unmoderated Usability Testing | Evaluation[48] | Test participants perform the evaluation on their own without the presence of an evaluator.[48] The software directs the instructions straight to the test participants.[48] | – | "In the field of Remote User Testing, Unmoderated Usability Tests have lost some of their importance. Synchronous Usability Testing has replaced on-site tests."<br>"During Unmoderated Usability Testing, disruptions such as technical problems can occur."<br>"There are difficulties and no possibility to check whether the test participants performed or understood the tasks correctly." |

Applied Clinical Informatics © 2022. The Author(s).

**Supplementary Appendix Table C** (*Continued*)

| Not recommended eHealth usability evaluation method | Stage of software lifecycle | Description of not recommended eHealth usability evaluation method | Equivalent terms | Quotes |
|---|---|---|---|---|
| | | | | "The application of Unmoderated Usability Testing is especially not recommended in health care if the results must be reliable to achieve certifications, for instance." |
| Questionnaires | Evaluation[35] | Questionnaires are used to quantify subjective requirements of a software.[80] An example of a traditional questionnaire is the System Usability Scale.[80] | – | "I would definitely not use Questionnaires because many test participants who are representative of prospective users are needed. To obtain reliable statistical results, at least 30 test participants or more are needed." "I would not recommend using Questionnaires to evaluate eHealth systems rapidly because many of the scoring systems are cumbersome and too complex to understand quickly." |
| Card Sorting as well as Storyboard | Requirements[78] | Via Card Sorting, terms are categorized to evaluate the usability of a software,[78] such as an eHealth system. A Storyboard consists of defining personas to determine the requirements of a target group and use cases.[81] | – | I would avoid applying Card Sorting and Storyboard to evaluate eHealth systems. First, these methods origin from user experience and are not suitable to evaluate eHealth systems. Second, regarding Card Sorting the analysis of the findings becomes time-consuming, especially if there are many test participants. And, concerning Storyboard: If the presentation is performed extensively and in detail, the effort is likewise quite high. Since eHealth systems are becoming increasingly complex, I don't think that applying Card Sorting or Storyboard is suitable if there are time constraints in advance. |
| Shadowing | Evaluation[82] | A video recording made during the usability evaluation is subsequently reviewed by both the evaluator and the test participant to analyze the behavior of the test participant.[82] Thus, the usability evaluation requires twice as much time.[82] | – | "Shadowing is usually quite time-consuming because I need different situations from everyday use that belong more to user research. Not really an eHealth evaluation method suitable to perform an evaluation quickly anyway." |
| Persona-Based Inspection | Evaluation[68] | Experts put themselves in the shoes of Personas and evaluate the software based on the characteristics assigned to the Personas.[68] | – | "Since I have to keep the effort low, I would also not set up a Persona-Based Inspection because as far as both usability experts and application experts are available, from the respective medical domain, the implementation of an Expert Review (Walkthrough or Talkthrough) is much more appropriate." |

(*Continued*)

Applied Clinical Informatics © 2022. The Author(s).

**Supplementary Appendix Table C** (*Continued*)

| Not recommended eHealth usability evaluation method | Stage of software lifecycle | Description of not recommended eHealth usability evaluation method | Equivalent terms | Quotes |
|---|---|---|---|---|
| Think Aloud combined with Questionnaire | Requirements,[80] Evaluation,[35] Design, and Implementation[24] | During Think Aloud, test participants who are representative of prospective users perform tasks by expressing their thoughts aloud.[24] Questionnaires are used to quantify subjective requirements of a software.[80] An example of a traditional questionnaire is the System Usability Scale.[80] | Further information about equivalent terms of Think Aloud can be found in ►**Appendix Table B**. | "Basically, I don't think it makes sense to plan a mix of eHealth usability evaluation methods, like a combination of Think Aloud and Questionnaire, from the beginning. If one method, like Think Aloud, was performed, I would wait and observe if the results were satisfactory. Considering low resources, a standardized questionnaire can be carried out quickly, but the question is how reliable the results are with a small number of test participants." |
| Co-Discovery Evaluation | Early design phase[21] | Test participants perform tasks simultaneously on the same technology,[21] such as eHealth systems. This can also be considered constructive user interaction,[83] which does not result in individual data[79] but in high-quality collective data.[30] | Paired-User Testing,[21] Co-Discovery Learning,[83] Constructive Interaction,[83] Co-Discovery,[30] Co-Discovery Methods[79] | "I would not recommend Co-Discovery Evaluation because in my experience it is associated with an insane amount of work in advance. Co-Discovery Evaluation is useful if an evaluation is to be conducted with several groups of test participants in parallel to compare results. Many groups, each with two test participants, must of course be supervised, which is very time-consuming." |
| Field Observation | Requirements[35] | The usability evaluation is not conducted in a usability laboratory but in a real environment,[84] such as the workplace[85] or the test participant's home environment.[64] | Field Study/User Observation,[85] In-Situ Usability Testing,[84] Portable Evaluations[64] | "If you do Field Observation properly, you need a relatively large amount of time and money. It is not enough to observe one test participant; you have to observe several test participants over a long period of time and if you don't have enough time, it's a bit difficult."<br>"So with Field Observation, it always involves a lot of effort, no matter how many test participants you have or not."<br>"You usually need a certain amount of support for the evaluation and so on. I would say that this is difficult and not recommendable from a time-critical point of view." |
| Online Survey | Evaluation | Questionnaire evaluation with a large number of test participants to determine quantitative results. | – | "I would avoid implementing an Online Survey in the medical domain if time resources are tight because a high number of test participants are required, which are difficult to recruit." |

Note: Not recommended eHealth usability evaluation methods are ordered by the number of experts' choice.

Applied Clinical Informatics © 2022. The Author(s).